Decision Trees using ID3

Engr. Dr. Jalees Ur Rahman

Faculty Offices CS – M Block
jalees.urrahman@cust.edu.pk



Introduction to Decision Trees

- Supervised learning used for both classification & regression.
- Consists of root node, branches, internal nodes & and leaf nodes.
- Internal nodes are attribute tests
- Branches are attribute values
- Leaf nodes are final decisions

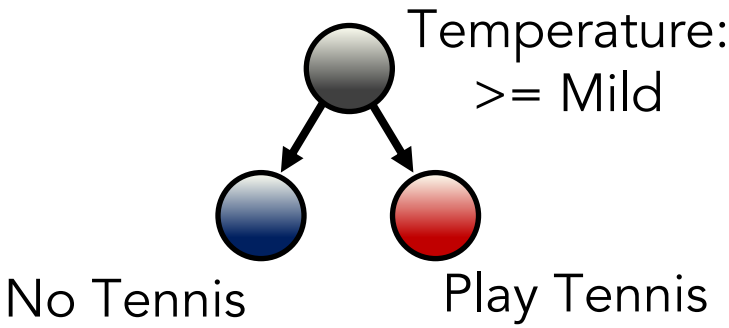
Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Introduction to Decision Trees

- Want to predict whether to play tennis based on temperature, humidity, wind, outlook

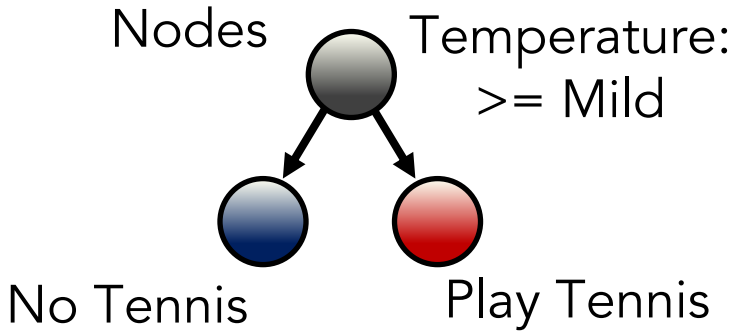
Introduction to Decision Trees

- Want to predict whether to play tennis based on temperature, humidity, wind, outlook
- Segment data based on features to predict result



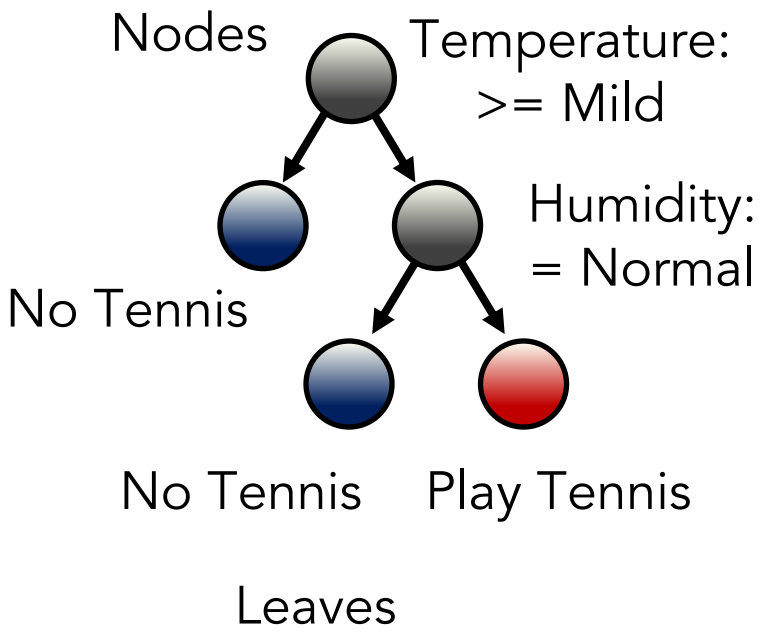
Introduction to Decision Trees

- Want to predict whether to play tennis based on temperature, humidity, wind, outlook
- Segment data based on features to predict result



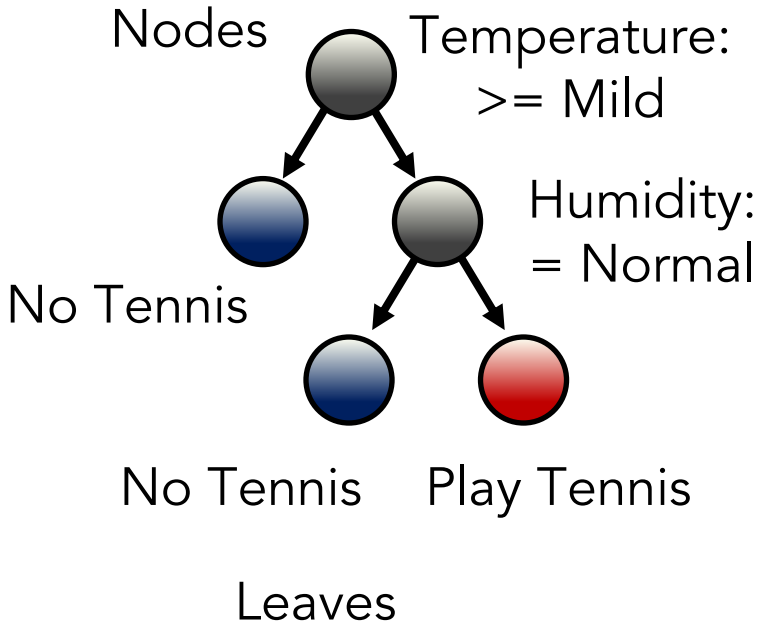
Introduction to Decision Trees

- Want to predict whether to play tennis based on temperature, humidity, wind, outlook
- Segment data based on features to predict result

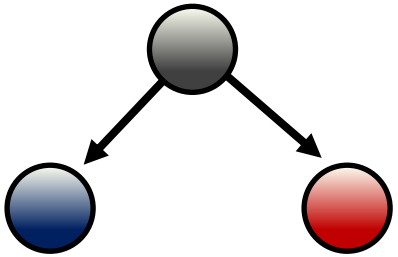


Introduction to Decision Trees

- Want to predict whether to play tennis based on temperature, humidity, wind, outlook
- Segment data based on features to predict result
- Trees that predict categorical results are **decision trees**

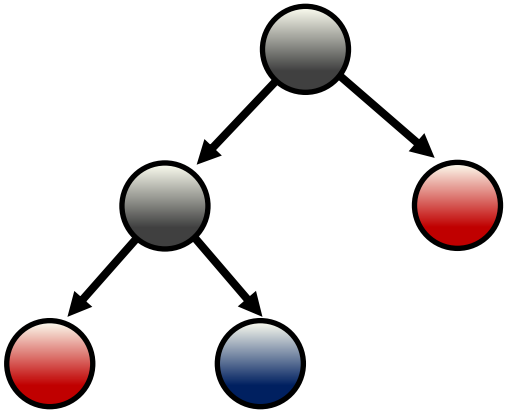


Building a Decision Tree



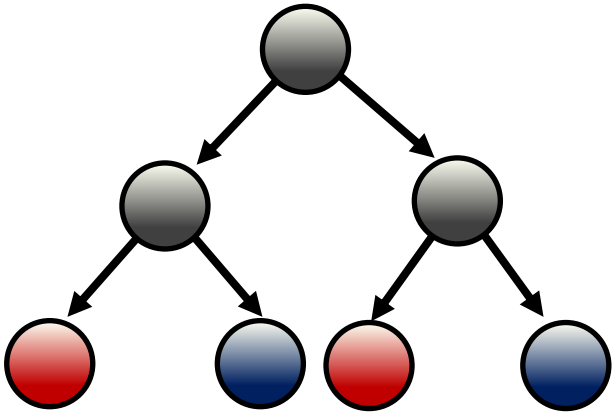
- Select a feature and split data into binary tree

Building a Decision Tree



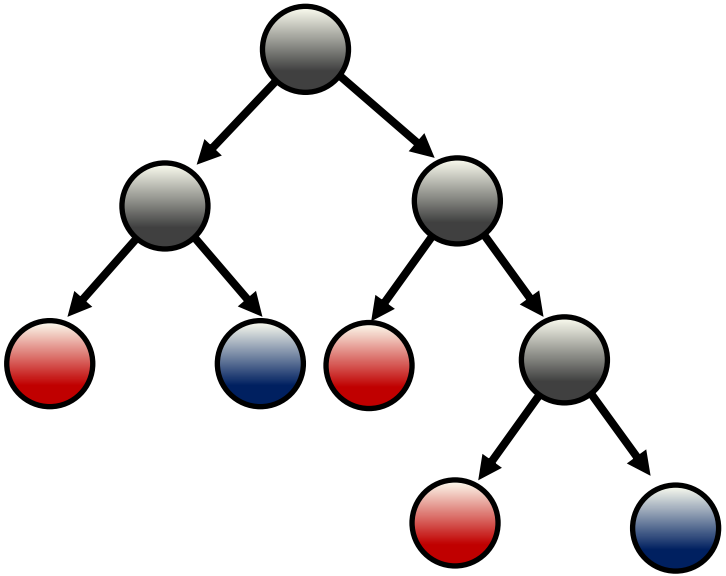
- Select a feature and split data into binary tree
- Continue splitting with available features

Building a Decision Tree

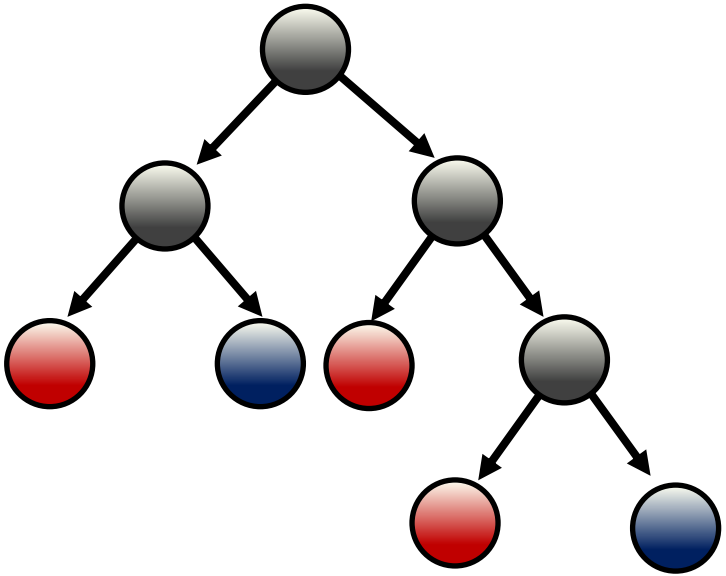


- Select a feature and split data into binary tree
- Continue splitting with available features

How Long to Keep Splitting?



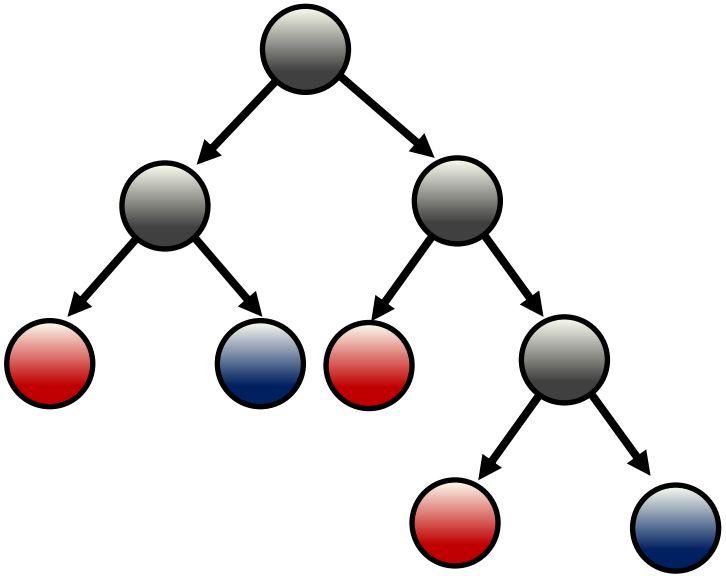
How Long to Keep Splitting?



Until:

- Leaf node(s) are pure (only one class remains)

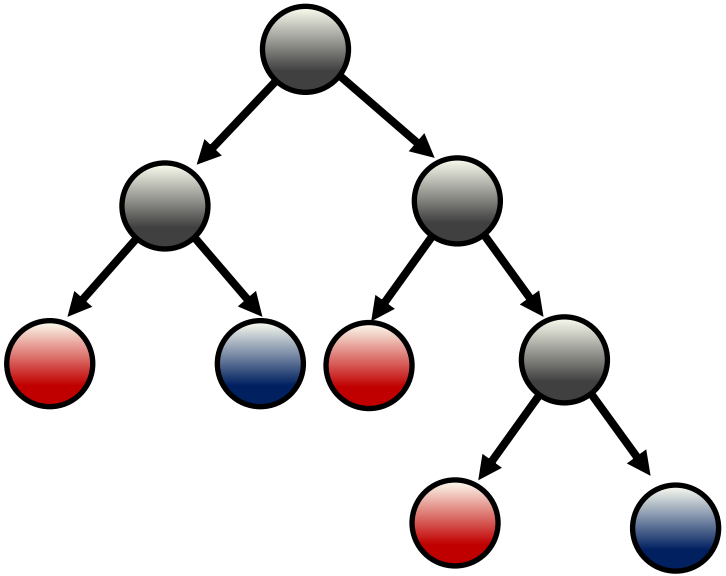
How Long to Keep Splitting?



Until:

- Leaf node(s) are pure (only one class remains)
- A maximum depth is reached

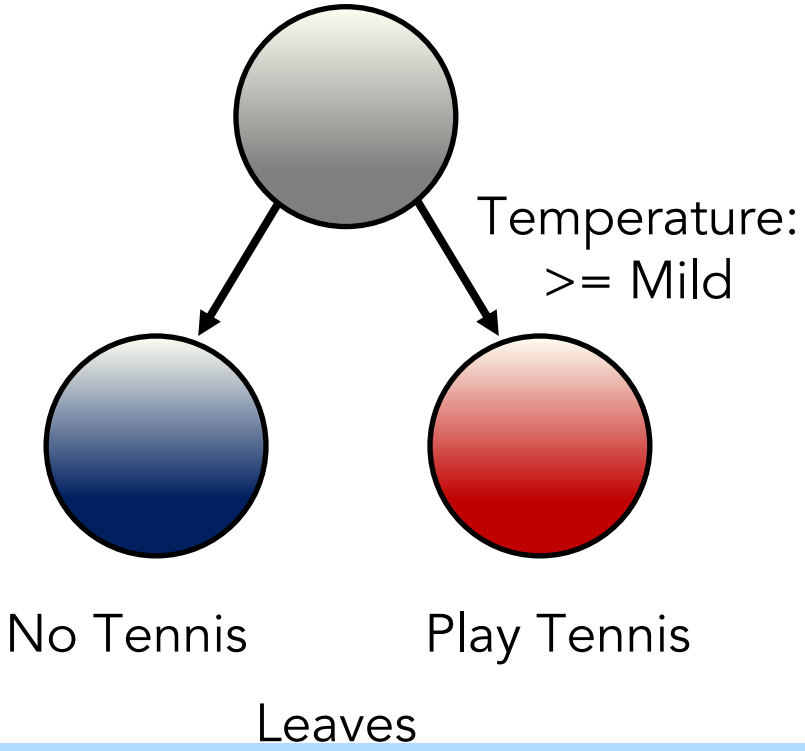
How Long to Keep Splitting?



Until:

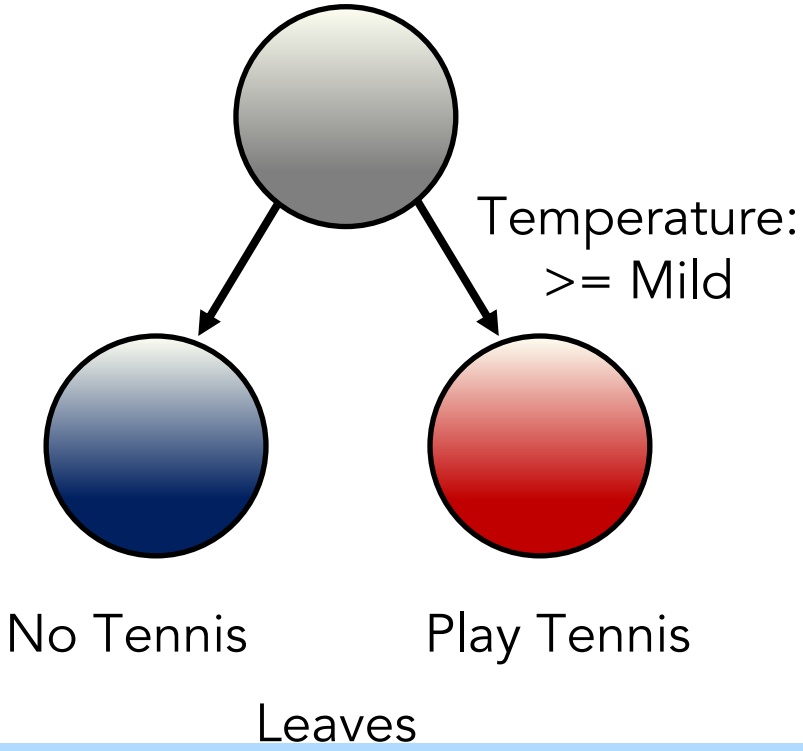
- Leaf node(s) are pure (only one class remains)
- A maximum depth is reached
- A performance metric is achieved

Building the Best Decision Tree



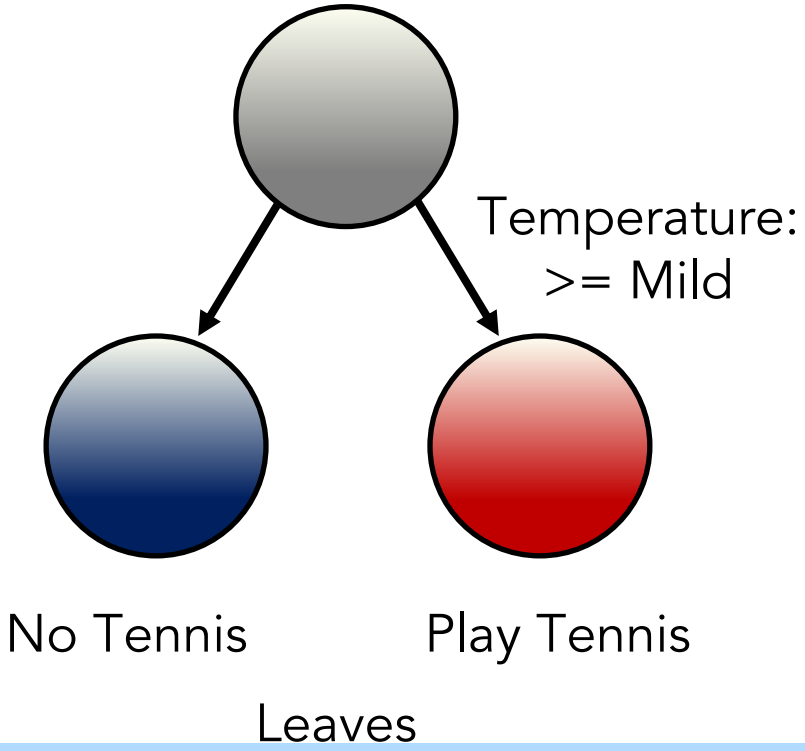
- Use greedy search: find the best split at each step

Building the Best Decision Tree



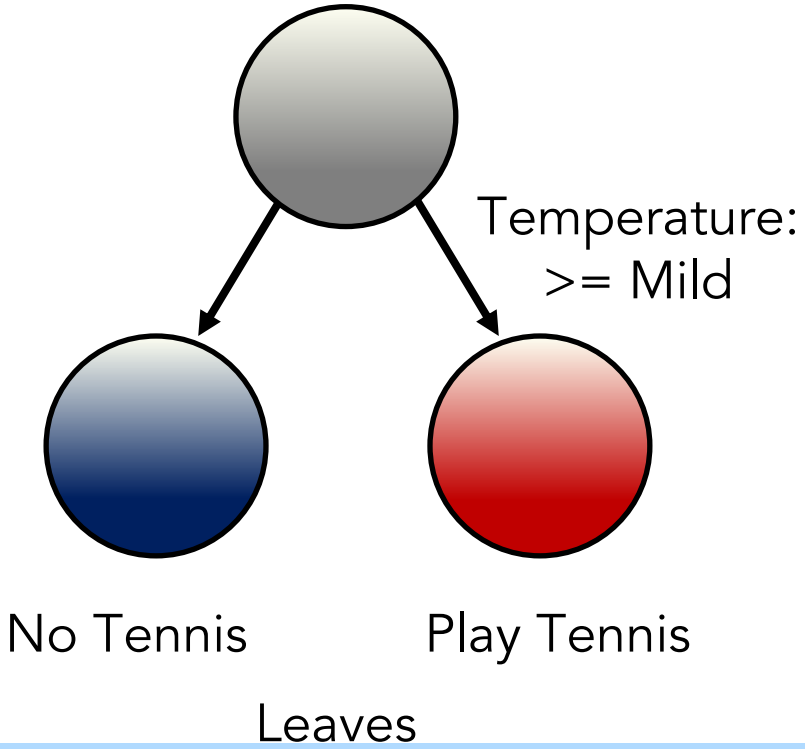
- Use greedy search: find the best split at each step
- What defines the best split?

Building the Best Decision Tree



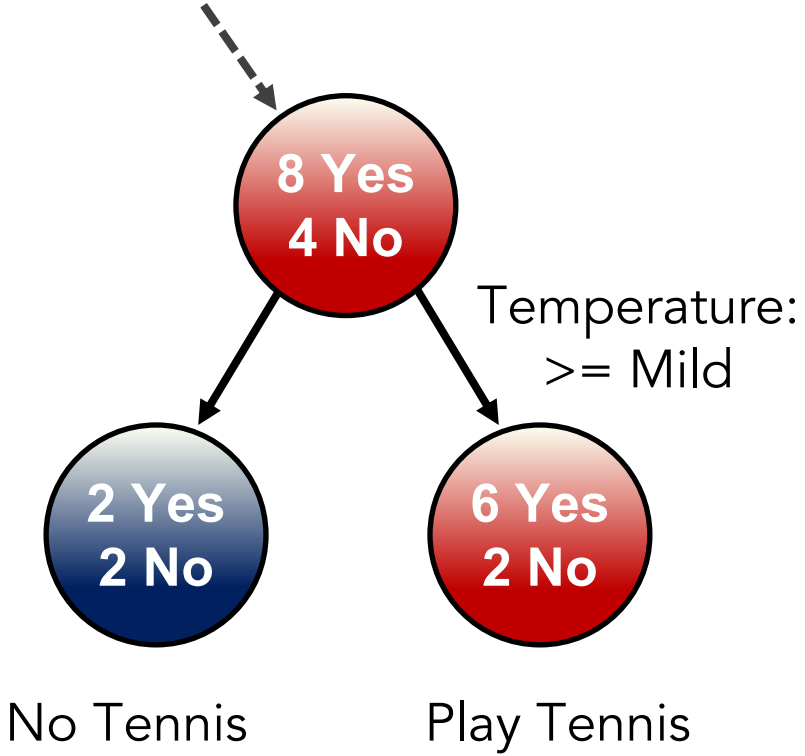
- Use greedy search: find the best split at each step
- What defines the best split?
- One that maximizes the information gained from the split

Building the Best Decision Tree



- Use greedy search: find the best split at each step
- What defines the best split?
- One that maximizes the information gained from the split
- How is information gain defined?

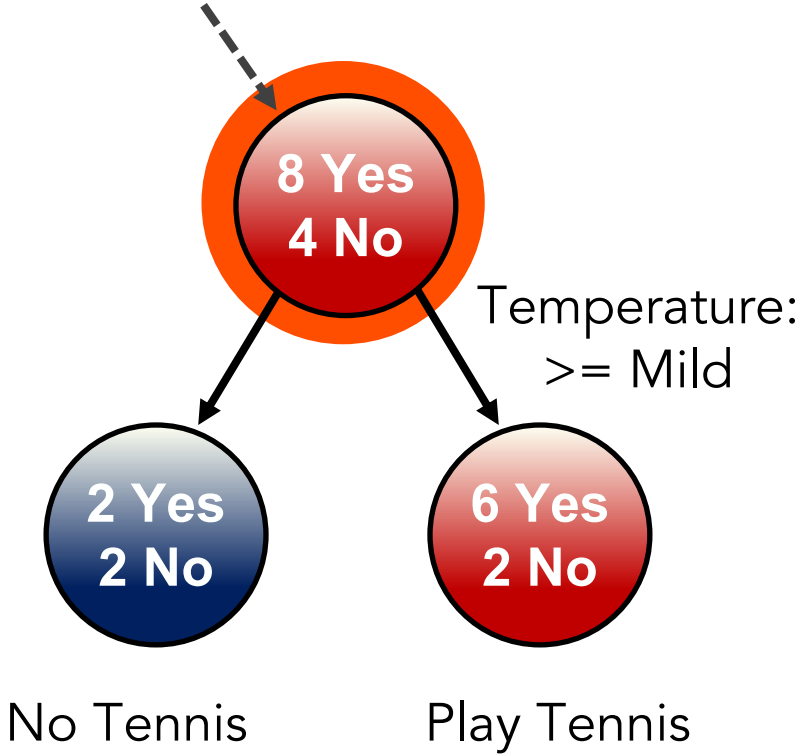
Splitting Based on Entropy



Entropy Equation

$$H(S) = -\sum_{i=1}^{n} p(i|t) log_2[p(i|t)]$$

Splitting Based on Entropy



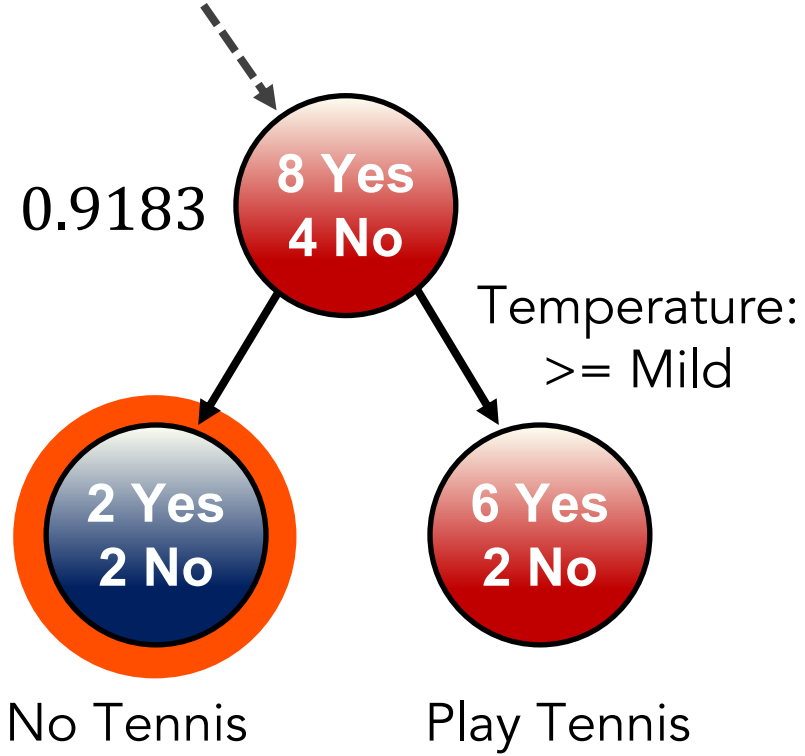
Entropy Equation

$$H(S) = -\sum_{i=1}^{n} p(i|t) log_2[p(i|t)]$$

Entropy Before

$$-{}^{8}/_{12}\, log_2({}^{8}/_{12}) \; -{}^{4}/_{12}\, log_2({}^{4}/_{12}) = 0.9183$$

Splitting Based on Entropy



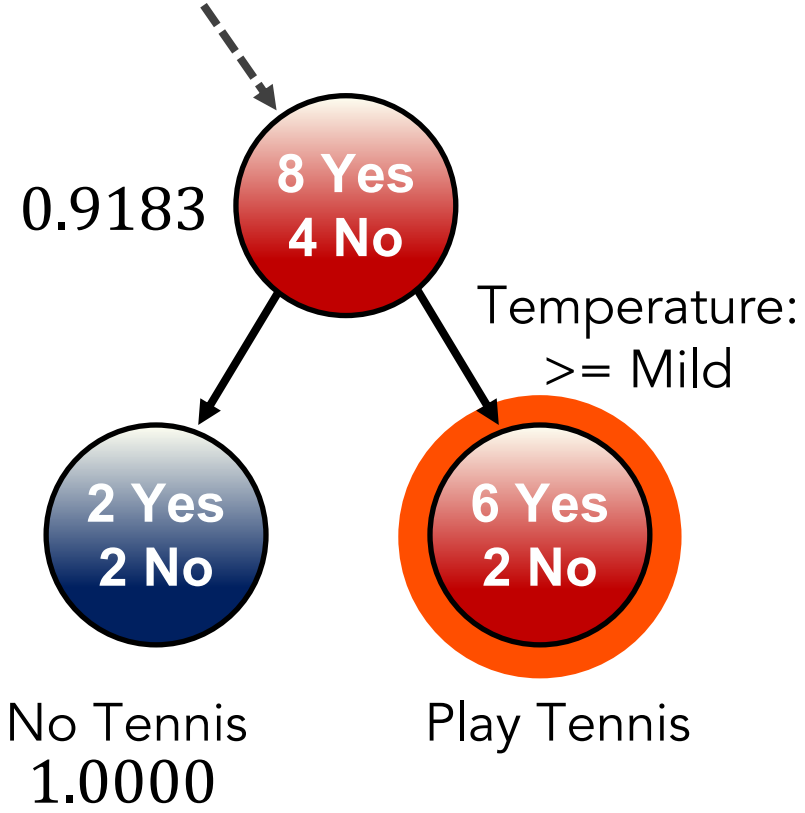
Entropy Equation

$$H(S) = -\sum_{i=1}^{n} p(i|t) log_2[p(i|t)]$$

Entropy Left Side

$$-{}^{2}/_{4}\, log_2({}^{2}/_{4}) \; -{}^{2}/_{4}\, log_2({}^{2}/_{4}) = 1.0000$$

Splitting Based on Entropy



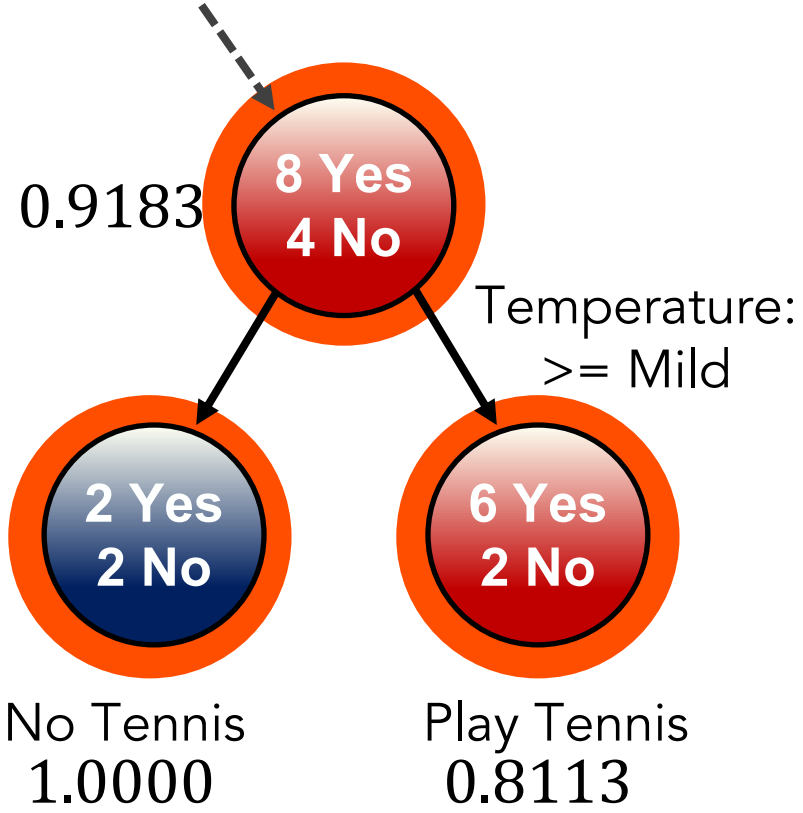
Entropy Equation

$$H(S) = -\sum_{i=1}^{n} p(i|t) log_2[p(i|t)]$$

Entropy Right Side

$$-{}^{6}/_{8}\, log_2({}^{6}/_{8}) \; -{}^{2}/_{8}\, log_2({}^{2}/_{8}) = 0.8113$$

Splitting Based on Entropy



Entropy Equation

$$H(S) = -\sum_{i=1}^{n} p(i|t) log_2[p(i|t)]$$

Entropy Change

$$0.9183 - {}^{4}/_{12} * 1.0000 - {}^{8}/_{12} * 0.8113$$
$$= 0.0441$$

ID3 Algorithm for Decision Trees

- Iterative Dichotomiser 3
- **Core Concept:** To decide which attribute to split.
- **Entropy:** Measure of the amount of uncertainty or randomness.
- **Information Gain:** Measure of the reduction in entropy after a dataset is split on a specific attribute.

$$IG(S, A) = H(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} H(S_v)$$

- **Steps:**
- Start with the entropy of the full dataset.
- Find the weighted entropy after splitting on an attribute.
- Subtract the weighted entropy from the original entropy.

Example Solution

$$H(S) = -\sum_{i=1}^{n} p(i|t) log_2[p(i|t)]$$

- First calculate entropy of data

$$H(S) = -\frac{9}{14}\log_2\left(\frac{9}{14}\right) - \frac{5}{14}\log_2\left(\frac{5}{14}\right)$$

$$H(S) = -0.643\log_2(0.643) - 0.357$$

$$H(S) = 0.940$$

- We have four attributes to select:

- Outlook, Temperature, Humidity Wind

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Example Solution

$$H(Sunny) = -\frac{2}{5}\log_2\left(\frac{2}{5}\right) - \frac{3}{5}\log_2\left(\frac{3}{5}\right)$$

$$H(Sunny) = 0.971$$

$$H(Rain) = -\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right)$$

$$H(Rain) = 0.971$$

$$H(Overcast) = -\frac{4}{4}\log_2\left(\frac{4}{4}\right)$$

$$H(Overcast) = 0$$

$$H(S \mid Outlook) = \frac{5}{14}(0.971) + \frac{4}{14}(0) + \frac{5}{14}(0.971)$$

$$H(S \mid Outlook) = 0.694$$

$$IG(S, Outlook) = 0.940 - 0.694$$

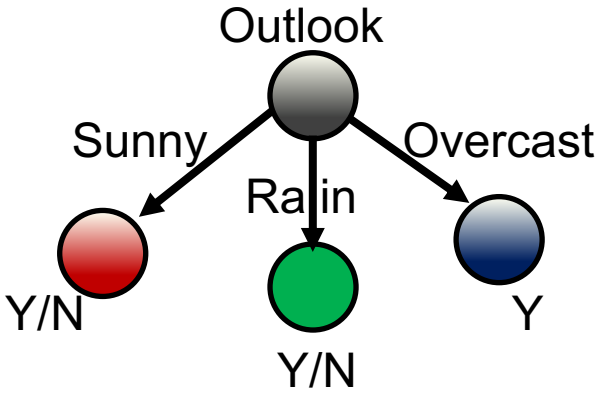
$$IG(S, Outlook) = 0.246$$

Humidity	0.151
Wind	0.048
Temperature	0.029

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Root Note: Outlook

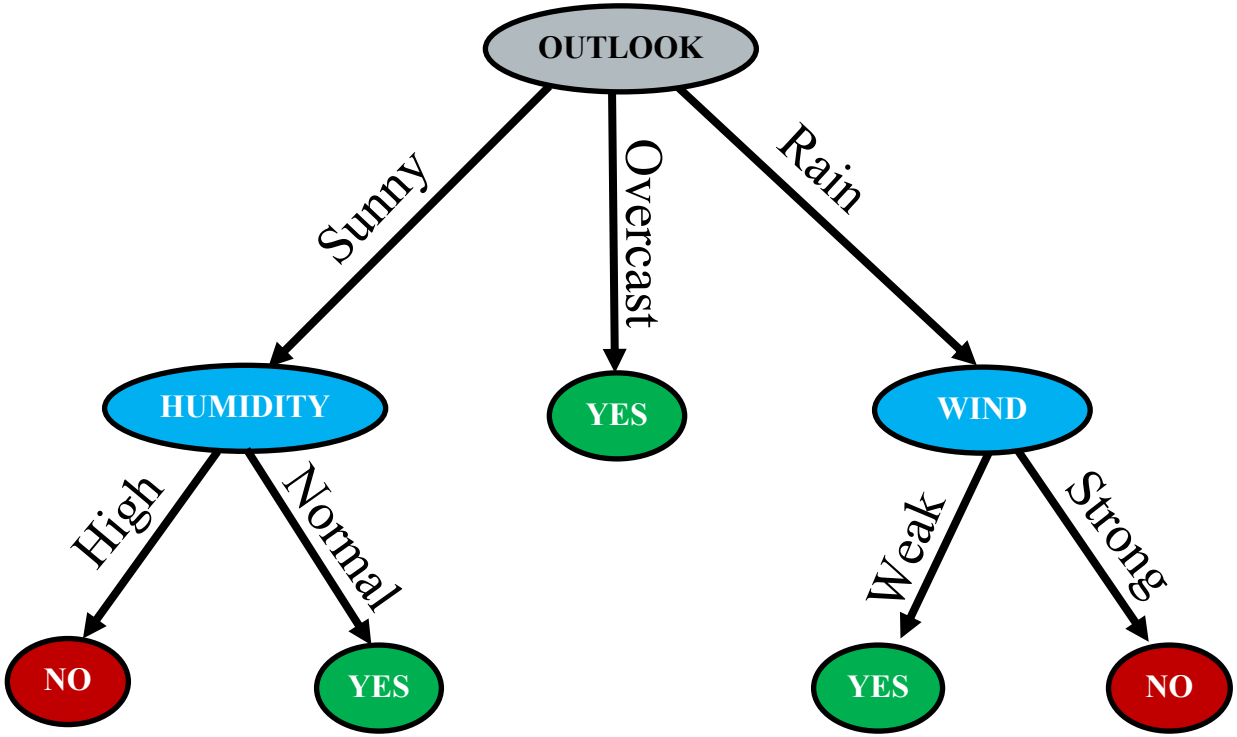
- Overcast has Yes in all cases, no need to explore
- Sunny and Rain need to be explored
- Dataset for Sunny
- Dataset for Rain



Day	Temperature	Humidity	Wind	PlayTennis
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

Day	Temperature	Humidity	Wind	PlayTennis
D4	Mild	High	Weak	Yes
D5	Cool	Normal	Weak	Yes
D6	Cool	Normal	Strong	No
D10	Mild	Normal	Weak	Yes
D14	Mild	High	Strong	No

Final Tree using ID3 Algorithm



Examples to think

If Outlook = Overcast, then PlayTennis = ?

If Outlook = Sunny and Humidity = High, then PlayTennis = ?

If Outlook = Sunny and Humidity = Normal, then PlayTennis = ?

If Outlook = Rain and Wind = Weak, then PlayTennis = ?

If Outlook = Rain and Wind = Strong, then PlayTennis = ?


Thank
You!